ITA03 MOBILE COMPUTING

Assignment 3

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Executive summary and modelling assumptions

This submission answers all seven questions in the ITA03 Assignment 3 brief. The recommended emergency design combines a local multi-hop responder mesh, AODV-style on-demand routing, and Mobile IPv4 at a gateway when a path to the home agent exists. Local service continues when public infrastructure fails; Mobile IP preserves address reachability only while a route to the registered care-of address is available.

Numerical cases are illustrative hand calculations from stated assumptions, not field measurements. No site map, traffic trace, radio standard, or measured channel data was supplied. Replace the illustrative independent-proposal placeholders in Q7 with the students' authentic pre-discussion notes; they are not evidence of work independently completed.

Assumption	Illustrative value	Purpose
Handoff interruption T_h	0.20 s	Break-before-make outage
Packet interval T_p	0.10 s	10 packets/s
Speed cases	1.5, 10, 20 m/s	Walking, vehicle, fast vehicle
Mobile IP registration	100+80+10 ms	Discovery + signalling RTT + processing
Outer IPv4 header	20 bytes	Tunnel overhead; no options
Q7 traffic / test	8 flows / 120 s	Suggested experiment plan

Q1 Wireless network architecture and mobility analysis

1.1 Architecture and communication path

A wireless access network contains mobile nodes (MNs), radio access points or base stations, backhaul, and network services. An MN associates with an access point; the access point bridges radio frames to the distribution network. In cellular service, the base

station and core also handle authentication, scheduling, mobility control, and inter-cell handoff. Radio capacity is finite and affected by interference, fading, contention, and changing rate.

Mobile user / MN ⇄ AP or base station ⇄ backhaul / core ⇄ application server

radio link access control routing / services

Component	Role	Constraint
Mobile node	Creates and receives application packets	Battery, radio range, attachment changes
AP / base station	Radio access and forwarding	Capacity, coverage boundary, outages
Backhaul/core	Connects access sites to services	Congestion, delay, infrastructure reliance
Wireless link	Shared radio transmission	Interference, fading, contention, finite bandwidth

1.2 Mobility sequence and handoff logic

A handoff is initiated when the serving-link quality falls below a release threshold and a candidate remains above an admission threshold for a dwell period. Hysteresis prevents ping-pong. The MN detects the candidate, authenticates and associates, obtains a new address if required, updates location, switches forwarding, and resumes queued traffic. Make-before-break can reduce interruption when coverage overlaps.

ALGORITHM MobilityHandoff(MN, CurrentAP, CandidateAP)

INPUT: link quality, thresholds, hysteresis, candidate availability

OUTPUT: HANDOFF_SUCCESS, STAY, or HANDOFF_FAILURE

BEGIN

IF Candidate unavailable OR Quality(Candidate) < AdmissionThreshold THEN RETURN STAY

IF Quality(Current) > ReleaseThreshold AND Candidate advantage < Hysteresis THEN RETURN STAY

Authenticate and associate MN; obtain address / care-of address if required

Buffer delay-tolerant packets; preserve sequence numbers

IF association and location update succeed THEN switch path; flush queue; RETURN SUCCESS

ELSE retain old AP if usable; otherwise mark disconnected; RETURN FAILURE

END

1.3 Worked interruption estimate

With packets every $T_p=0.10$ s and handoff interruption $T_h=0.20$ s, about $T_h/T_p=2$ packets are generated during one outage. For n handoffs in duration T, the exposed fraction under uniform traffic is $U=\min(1,nT_h/T)$. For coverage spacing $L=500$ m over $T=60$ s, expected crossings $n\approx vT/L$:

Speed	Distance/60 s	Expected crossings	Outage fraction	Meaning
1.5 m/s	90 m	0.18	0.06%	Likely no crossing
10 m/s	600 m	1.20	0.40%	About one handoff
20 m/s	1,200 m	2.40	0.80%	About two or three

The reproducible corridor model in the companion Python file uses a fixed 37 ms packet phase, no buffering, and 60 seconds. It sent 600 packets in each run. Results: 1.5 m/s delivered 600/600 with no boundary crossing; 10 m/s delivered 598/600 (99.67%) with one crossing; 20 m/s delivered 596/600 (99.33%) with two crossings. The deterministic outcomes depend on packet phase; expected values above remain averages. Buffering can prevent loss but adds queueing delay; higher speed increases handoff frequency.

Case	Behavior	Record
Stable attachment	Baseline access and backhaul delay	Sent/delivered; mean and 95th percentile delay
Successful handoff	Brief pause; buffering may reorder packets	Handoff duration, loss, reordering

Coverage hole	No usable AP; queue or drop	Outage duration and queue depth
Repeated mobility	More signaling and interruption	Handoffs/min and delivery ratio

Q2 Mobile IP and mobility management

2.1 Entities and packet delivery

Entity	Function
Mobile Node (MN)	Keeps a stable home address while visiting another network.
Home Agent (HA)	Stores home-address to care-of-address binding and tunnels packets.
Foreign Agent (FA)	In classic Mobile IPv4, advertises visitor service and may provide CoA / decapsulation.
Correspondent Node (CN)	Peer sending packets to the MN's home address.

The MN discovers mobility agents, obtains a care-of address (CoA), and sends an authenticated registration request through the FA to the HA. The HA installs a time-limited binding. Packets addressed to the home address reach the HA, which encapsulates them to the CoA; the tunnel endpoint decapsulates and delivers them. The CN-to-HA-to-MN path is triangular when direct delivery to the visited network is possible.

CN → packet for home address → HA: lookup binding[HomeAddress]=CoA

HA: encapsulate original packet to CoA → FA / endpoint → decapsulate → MN

MN return traffic may go directly to CN or through HA, by policy

2.2 Registration pseudocode

ALGORITHM RegisterAndTunnel(MN, HA, FA, CoA)

INPUT: home address, CoA, lifetime, authenticated registration data

OUTPUT: binding status and delivery decision

BEGIN

Discover agent; obtain CoA; create request(HomeAddress, CoA, Lifetime, Identifier)

Authenticate; SEND request via FA TO HA

IF HA authenticates and authorizes THEN store binding until expiry; reply ACCEPTED

ELSE reply REJECTED; RETURN REGISTRATION_FAILURE

WAIT for authenticated reply; retry only within configured limit on timeout

IF accepted THEN FOR each packet to HomeAddress: encapsulate to CoA and forward

IF tunnel receives packet THEN decapsulate and DELIVER

ELSE queue/retransmit if policy permits; record failure

RETURN REGISTRATION_SUCCESS

END

2.3 Delay, overhead and triangular routing

Illustrative registration estimate: $T_{reg} \approx T_{discovery} + RTT(MN/FA \leftrightarrow HA) + T_{processing} = 100 + 80 + 10 = 190$ ms. Actual delay depends on advertisement timing, packet loss, timers, and RTT. A lost exchange may add timeout delay; rapid movement can leave stale bindings or cause signaling bursts.

For a 1,500-byte original IP datagram, a 20-byte outer IPv4 header produces a 1,520-byte tunnel packet. Overhead/original = 20/1500 = 1.33%; original bytes as a fraction of encapsulated datagram = 1500/1520 = 98.68%. Link headers and tunnel extensions are excluded. Path MTU can require smaller packets or fragmentation.

Path	Illustrative one-way delay	Calculation	Effect
Triangular CN→HA→MN	60 ms	35+25 ms	Indirect but reachable
Direct optimized path	40 ms	Direct route	20 ms less, a 33% reduction vs 60 ms
Registration exchange	190 ms	100+80+10 ms	Can dominate short flows

2.4 Challenges and mitigation

Challenge	Effect	Response

Registration delay	Pause; packets head to old location	Fast detection, buffering, controlled timers, make-before-break
Tunnel overhead	More bytes and processing; MTU risk	Account for encapsulation and path MTU
Triangular route	Longer path and HA load	Secure route optimization or well-placed HA
Handoff loss	Old path fails before new binding	Buffer/retry and application acknowledgements
Security	Forged binding redirects traffic	Authenticate registration; replay defense; key management
Scalability	HA state/signaling grow with moving users	Regional hierarchy, capacity, rate control

Frequent changes may cause stale bindings, signaling bursts, reordering and loss. Hysteresis, buffering, minimizing unnecessary registrations, overlapping paths, and secure authenticated updates reduce these effects. Mobile IP cannot bridge a total partition: it provides location-based forwarding only when the CoA is reachable.

Q3 Cellular networks versus ad hoc networks

CELLULAR: MN ⇄ base station ⇄ operator core ⇄ service

AD HOC: MN ⇄ MN ⇄ MN ⇄ gateway/service (multi-hop; peers also forward)

Parameter	Cellular	Ad hoc
Infrastructure	Planned base stations, backhaul and core	No fixed infrastructure needed for local peer communication
Communication	Device to base station then core	Direct or distributed multi-hop
Routing	Operator-managed	Distributed discovery and maintenance

Mobility	Managed handoff between cells	Dynamic topology; any hop may break
Topology	Engineered coverage	Decentralized changing graph
Scalability	High with engineered spectrum/capacity	Constrained by contention, energy and routing overhead
Handoff	Network-assisted	Distributed attachment/route repair
Failure	Site/core failure can remove broad service	Alternate paths may exist; sparse networks partition
Applications	Public wide-area voice/data	Disaster, military, temporary field networks

When fixed infrastructure is unavailable, choose an ad hoc mesh for local responder communication because it can deploy rapidly and forward peer-to-peer without a working base-station core. Use surviving cellular or deployable cells as extra gateways where available. A partitioned mesh continues locally within each component but cannot cross the partition until connectivity returns.

Q4 Mobile IP application and routing requirements

4.1 Scenario: ambulance-to-hospital telehealth

An ambulance sends vitals and ECG snapshots to a hospital endpoint while crossing network boundaries. Entities include ambulance terminal (MN), paramedic handhelds, access sites, HA, optional FA/CoA, hospital server (CN), and dispatch. Mobile IP gives the terminal a stable home address across visited IP networks.

Vitals / ECG terminal (MN) ⇄ cellular or Wi-Fi access ⇄ visited network

| |

home address FA / CoA

| ⇅ tunnel

Home Agent (HA) ⇄ hospital / dispatch server (CN)

Requirement	Design implication

Latency	Prioritize alarms and small vitals; use local alarm rules if WAN is slow.
Delivery	Acknowledge and retry critical records; adapt video bitrate.
Mobility	Stable identity and rapid registration; avoid signal-fluctuation updates.
Connectivity	Make-before-break where possible; queue nonurgent records.
Security	Mutual authentication, encryption, least privilege, audit trail.
Resilience	Dual radio/backhaul when practical; local display and alarms during outage.

Mobile IP suits stable reachability across subnets when the HA is reachable. It does not supply coverage, application encryption, clinical reliability, or real-time guarantees. Separate critical telemetry from bulk imagery and avoid queueing delays beyond the clinical tolerance.

4.2 Message flow

1. MN detects visited network and obtains CoA.

2. MN authenticates registration to HA; HA updates binding.

3. CN sends to home address; packet reaches HA.

4. HA tunnels to CoA; endpoint decapsulates and delivers.

5. On handoff, MN registers new CoA and queues/retries if the old tunnel fails.

Q5 Proactive routing with DSDV

5.1 Principle and route update

A proactive protocol maintains routes before traffic requests them. DSDV advertises destination, next hop, hop count and destination sequence number. Newer destination sequence is preferred; for equal sequence values, fewer hops wins. A broken route is advertised unreachable with a newer sequence value. Continuous readiness reduces first-packet discovery delay but periodic advertisements consume radio capacity.

ALGORITHM ProcessDSDVUpdate(Node, AdvertisedRoute)

INPUT: Destination, NextHop, HopCount, DestinationSequence, route table

OUTPUT: updated entry or unchanged table

BEGIN

CandidateHop ← AdvertisedRoute.HopCount + 1

IF destination absent OR received sequence > stored sequence THEN install candidate

ELSE IF sequences equal AND CandidateHop < stored hops THEN install candidate

ELSE keep current route

IF neighbor link DOWN THEN invalidate affected routes and advertise unreachable

with the next sequence value (an odd value marks the invalid route)

RETURN route table

END

5.2 Route trace

Consider A—B—C—D with a surviving alternate path A—E—D. Initially A routes to D via B, three hops, destination sequence 8. After B–C fails, B invalidates the old route; a fresh sequence-10 route through A–E–D is learned. The exact convergence time depends on advertisement propagation.

Node	Before: next hop / hops / seq	After update: next hop / hops / seq	Reason
A	B / 3 / 8	E / 2 / 10	Newer route via alternate
B	C / 2 / 8	A / 3 / 10	Old next hop invalid; fresh route learned
C	D / 1 / 8	D / 1 / 10	Learns new destination sequence
D	self / 0 / 8	self / 0 / 10	Destination sequence advances

A route is not converged until updates reach affected nodes. During convergence packets may be dropped or use stale information. Sequence numbers and invalid-route updates reduce stale loops but do not eliminate transient loss.

5.3 Overhead and use cases

With N nodes, a full periodic update can carry O(N) entries per sender, giving roughly $O(N^2)$ advertised entries network-wide per update when every node broadcasts its table. Incremental updates reduce typical cost, but triggered advertisements still consume airtime. DSDV suits relatively stable networks with regular traffic and very low route-setup delay requirements. Larger networks and frequent movement increase update overhead and staleness.

Q6 Reactive routing with AODV and comparison

6.1 AODV discovery and maintenance

AODV discovers a route only on demand. A source without a valid route broadcasts a Route Request (RREQ) with source and request ID. Receivers suppress duplicates and record a reverse route to the source. The destination or eligible fresh route holder returns a Route Reply (RREP) along that reverse route; forwarding nodes install forward routes. On active-link failure, affected entries are invalidated and a Route Error (RERR) informs precursors; the source may rediscover.

ALGORITHM AODVDiscoverAndMaintain(Source, Destination)

INPUT: neighbors, route table, request ID, active packet queue

OUTPUT: VALID_ROUTE, ROUTE_NOT_FOUND, or ROUTE_REPAIRED

BEGIN

IF valid destination route and next-hop link UP THEN forward queued packets; RETURN

Increment RequestID; broadcast RREQ

Receiver discards duplicate, records reverse route, forwards once if needed

Destination/fresh route holder sends RREP along reverse route

IF RREP arrives before timeout THEN install route; send queued data; RETURN VALID

ELSE drop or retain by queue policy; RETURN ROUTE_NOT_FOUND

ON active next-hop failure: invalidate route; send RERR to precursors

IF source still has traffic THEN initiate bounded discovery

END

6.2 Trace and protocol comparison

Step	Event	Action	State
1	S needs D	No route; broadcast RREQ id=17	UNKNOWN
2	A receives	Record reverse next hop S; forward once	Reverse valid
3	B receives	Record reverse next hop A; forward	Reverse valid
4	D receives	Create RREP	Destination local
5	D→B→A→S	RREP installs forward route at each node	VALID
6	S sends data	Forward S→A→B→D	ACTIVE
7	A–B breaks	Invalidate; send RERR; rediscover if needed	INVALID / DISCOVERY

Dimension	DSDV proactive	AODV reactive
Acquisition	Routes continuously maintained	RREQ/RREP when needed
Idle overhead	Periodic table advertisements	Low without active demand; flood on discovery
First-packet delay	Low if converged	Includes route setup if no route
Mobility	Periodic/triggered updates may lag	RERR then repair/discovery on active break
State	Destination routes maintained	Active destination and reverse request state

Fit	Steady traffic/stable topology	Intermittent traffic/dynamic topology

For highly dynamic, intermittent responder traffic, choose AODV initially: it avoids continuous destination advertisements when no flow needs them. Its costs are route setup delay and RREQ broadcast overhead at flow start or repair. If topology stabilizes and traffic is continuous, DSDV may lower first-packet delay. If nearly every packet triggers a discovery, AODV flooding can be costly.

Q7 Collaborative mobile emergency network design

7.1 Stage 1: independent proposal records

The assignment requires authentic pre-discussion proposals. The samples below are placeholders, not evidence of actual independent brainstorming. Each student should replace the corresponding sample with their own dated note made before joint discussion.

Student	Illustrative sample placeholder — replace with authentic note
A	Deploy a local responder mesh. Keep stable home identity; use a reachable command-vehicle gateway as FA/CoA and a regional HA when backhaul exists. Retain local service if disconnected.
B	Use AODV for bursty responder traffic and moving links. Monitor active paths, send RERR on failure, rediscover with bounded retries, and use alternate gateways. Measure delivery, delay and control messages per delivered packet.

7.2 Compare proposals

Dimension	A emphasis	B emphasis	Joint interpretation
Agreement	Local mesh without fixed sites	On-demand multi-hop	Both support local communication
Focus	Identity/external reachability	Route discovery/repair	Mobile IP and path routing are distinct functions

Strength	Stable address and gateway	Traffic-driven overhead	Use Mobile IP only with HA path; AODV inside mesh
Risk	Tunnel cannot cross partition	RREQ flooding/first delay	Keep local service; limit retries and prioritize traffic
Measures	Registration and tunnel	Routing overhead and delay	Measure costs separately

7.3 Joint architecture, registration and forwarding

Regional command / hospital CN

⇅

Home Agent (HA)

⇅ secure backhaul

Ambulance MN ⇄ responder mesh ⇄ command vehicle gateway (FA / CoA)

⇅ ⇅ ⇅

medical team police node rescue / sensor nodes

Local mesh: multi-hop links and AODV; local services work without WAN.

Responder devices are mobile hosts and forwarding nodes. A gateway may have multiple radios and satellite, surviving cellular, or temporary backhaul. Mobile IPv4 operates at the external-network boundary when the HA is reachable; it binds a home address to the current CoA and tunnels external traffic. Local mesh traffic uses local addressing and AODV. Mobile IP does not create connectivity through a partition.

1. MN discovers a gateway and obtains local CoA.

2. If HA is reachable, authenticate registration; HA binds home address to CoA.

3. External CN sends to home address; HA tunnels to CoA; gateway forwards through mesh.

4. Local responder traffic routes directly by AODV.

5. On break, AODV invalidates route, sends RERR and searches alternate path/gateway.

6. If none exists, queue suitable records locally; external delivery waits for reconnection.

7. Re-register new CoA and synchronize queued records after reconnection.

7.4 Failure handling

Condition	Response
Single mesh link fails	RERR and rediscovery; use valid alternate next hop
Responder node fails	Expire neighbor and routes; preserve alternate gateway paths
Gateway loses backhaul	Keep local mesh; queue nonurgent records; indicate external service unavailable
Mesh partitions	Operate each connected component locally; carry/store-forward until paths return
Congestion	Prioritize distress, control and vitals; rate-limit imagery
HA unavailable	No confirmed new binding; retain local addressing and services
Security event	Authenticate nodes/registrations, encrypt data, revoke lost devices, audit changes

7.5 Analytical validation

Let T_h be radio handoff, T_reg registration, and T_route AODV discovery when no route is cached. A conservative sequential interruption is T_interrupt≈T_h+T_reg+T_route. Under illustrative values 0.20+0.19+0.15=0.54 s. At 10 packets/s, about 5.4 packets are exposed without buffering/recovery. Terms may overlap; actual discovery depends on hop count, contention and timers. For a 1,500-byte packet, the 20-byte tunnel header adds 1.33% at IP-datagram level.

Define PDR=received/generated×100%; mean delay=average(receive time−send time) for delivered packets; routing overhead=control transmissions/delivered packets. Report topology, radio range, traffic, seed and protocol implementation with these metrics.

7.6 Experimental plan and theory versus simulation

The companion Python file was executed for a reproducible abstract route-outage simulation, not a full radio or geometric network simulator. Each of eight flows generates

one packet/s for 120 s; mean path hops are ceil(sqrt(N)/2), with at least two; per-hop hazard is 0.1%, 0.5% and 1.2% per second for low, medium and high mobility. DSDV repairs at the next 5 s update; AODV discovers in 150 ms; packets wait up to 0.50 s. Data-hop delay is 10 ms. DSDV control transmissions are N per periodic update; AODV floods across N nodes and returns over the path on first discovery and each break. Fixed seed 7301, incremented by case, makes the result reproducible. This model isolates route break/recovery and control trade-offs; it omits radio interference, contention, geometry, MAC retries, and packet-level multi-hop forwarding. A full network simulator with multiple seeds is recommended for deployment claims.

Case	Nodes	Speed	Purpose
T1	12	1 m/s	Small, low-mobility baseline
T2	24	8 m/s	Moderate mobility and node count
T3	40	20 m/s	High-mobility stress case

Metric	Analytical expectation	Interpretation of a run
PDR	Usually declines with breaks/partitions; alternate paths help	Received divided by generated
Mean delay	AODV discovery delays first packet; converged DSDV may be faster	Separate setup and data-path delay
Control overhead	DSDV periodic baseline grows with network size; AODV grows with demand/breaks	Count control transmissions
Node count	More nodes add paths but also contention/state	Trend need not be monotonic
Mobility	More breaks raise repair/signaling and interruption	Use matched random seeds

The following are actual outputs from the stated abstract simulation, not measurements from a physical network. Each case generated 960 packets. A fixed deterministic seed is used for each case and protocol comparison; control-transmission counts follow the model rules above.

Protocol	Nodes	Mobility	PDR %	Mean delay ms	Control transmissions / delivered packet
AODV	12	Low	100.00	21.25	0.131
DSDV	12	Low	99.58	20.00	0.301
AODV	24	Medium	100.00	31.88	0.731
DSDV	24	Medium	96.67	30.00	0.621
AODV	40	High	100.00	41.79	2.292
DSDV	40	High	90.63	40.00	1.103

7.7 Final deployment recommendation

Deploy a multi-hop responder mesh with AODV for dynamic intermittent local traffic; add redundant gateways to surviving cellular, satellite or temporary backhaul; use Mobile IP at the boundary for stable external reachability. Prioritize emergency traffic, authenticate nodes and registrations, buffer carefully, and retain local services during WAN loss. Before deployment, survey terrain and radio links; test congestion, interference, density, battery life, delay objectives, partition recovery, security and interoperability. The design is unsuitable as the sole emergency path if node density cannot maintain connected components, radio capacity is inadequate, gateways cannot carry the load, or required delay bounds exceed best-effort mesh performance.

Contribution record and individual verification

The brief assigns primary responsibility by question, but both partners must understand the entire submission. Complete this factual contribution record after joint review.

Question	Student A primary	Student B primary	Joint work
Q1	Architecture/analysis	Mobility/performance	Scenario and interpretation

Q2	Mobile IP operation	Challenges	Architecture and mitigations
Q3	Cellular	Ad hoc	Comparison/disaster decision
Q4	Performance	Application design	Evaluation
Q5	DSDV design	Performance	Topology and route trace
Q6	AODV investigation	DSDV–AODV comparison	Protocol selection
Q7	Analytical validation	Experimental validation	Proposals and final design

Viva preparation prompts

Why does a care-of address change while the Mobile IP home address remains the stable identity?

Trace the reverse route and RREP in S–A–B–D.

If the HA is unreachable, what works locally and what external service does not?

How do AODV and DSDV overheads change as node count grows with intermittent traffic?

When might DSDV be preferable in a stable, continuously used network?